



2012 Half-Yearly Examination

FORM VI

MATHEMATICS 2 UNIT

Monday 20th February 2012

General Instructions

- Writing time — 2 hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Total — 90 Marks

- All questions may be attempted.

Section I – 10 Marks

- Questions 1–10 are of equal value.

Section II – 80 Marks

- Questions 11–15 are of equal value.
- All necessary working should be shown in every question.
- Start each question in a new booklet.

Collection

Section I Questions 1–10

- Place your multiple choice answer sheet inside the answer booklet for Question Eleven.

Section II Questions 11–15

- Start each of these questions in a new booklet.
- Write your candidate number clearly on each booklet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place the question paper inside your answer booklet for Question Eleven.

Checklist

- SGS booklets — 5 per boy
- Candidature — 80 boys

Examiner
JMR

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

Question One

Which of the following is equal to $e^x(e^x - \frac{1}{e})$?

- (A) $e^{x^2} - e$
- (B) $e^{2x} - e$
- (C) $e^{x^2} - e^{x+1}$
- (D) $e^{2x} - e^{x-1}$

Question Two

For the function $y = x^3 + 1$, which one of the following statements is true?

- (A) The function is odd.
- (B) The function is even.
- (C) The function is increasing for all values of $x > 0$.
- (D) There is a triple root at $x = -1$.

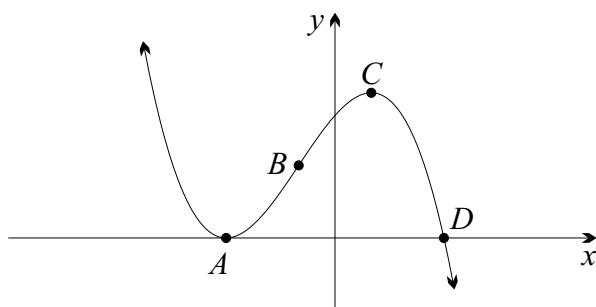
Question Three

The definite integral $\int_0^4 (x^2 + 2) dx$ is equal to

- (A) $\frac{88}{3}$
- (B) 18
- (C) $\frac{64}{3}$
- (D) $23\frac{1}{3}$

Exam continues next page ...

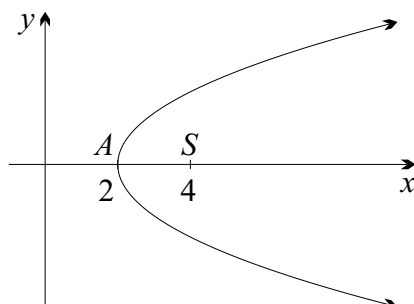
Question Four



Given the function $y = f(x)$ above, which of the following statements is false?

- (A) There is a local minimum at A .
- (B) The concavity changes at B .
- (C) There is a global maximum at C .
- (D) The zeroes occur at points A and D .

Question Five



For the parabola sketched above, point A is the vertex and point S is the focus. The equation of the parabola could be

- (A) $(y - 2)^2 = 8x$
- (B) $y^2 = 8(x - 2)$
- (C) $(y - 2)^2 = 8(x - 2)$
- (D) $(y + 2)^2 = 8(x - 2)$

Question Six

Which of the following is a primitive of $\frac{10}{x^2}$?

- (A) $-5x^2 + C$
- (B) $-10x^3 + C$
- (C) $-\frac{10}{x} + C$
- (D) $-\frac{10}{3x^2} + C$

Question Seven

The graph of the locus of the point $P(x, y)$ that moves so that its distance from a point $A(1, 1)$ is twice the distance from another point $B(4, 1)$ would be a

- (A) vertical line
- (B) parabola with a vertical axis of symmetry
- (C) parabola with a horizontal axis of symmetry
- (D) circle

Question Eight

The number of solutions to the equation $e^{x+1} + x^2 + 2 = 0$ may be found by sketching graphs. Which of the following statements is true?

- (A) We should sketch $y = e^{x+1} + 2$ and $y = -x^2$ to show there are no solutions.
- (B) We should sketch $y = x^2 + 2$ and $y = e^{x+1}$ to show there are two solutions.
- (C) We should sketch $y = e^{x+1}$ and $y = -x^2 - 2$ to show there are two solutions.
- (D) We should sketch $y = e^{x+1} + 2$ and $y = x^2$ to show only one solution.

Question Nine

The gradient of a line that is perpendicular to $3x + 5y - 5 = 0$ is

- (A) $-\frac{1}{3}$
- (B) $\frac{5}{3}$
- (C) $-\frac{5}{3}$
- (D) $-\frac{3}{5}$

Question Ten

The equation of a line with gradient $\frac{3}{2}$ and y -intercept $\frac{1}{2}$ is

(A) $y = 3(2x - 1)$

(B) $y = \frac{2x + 1}{3}$

(C) $3y = 1 - 2x$

(D) $y = \frac{1}{2}(3x + 1)$

_____ End of Section I _____

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

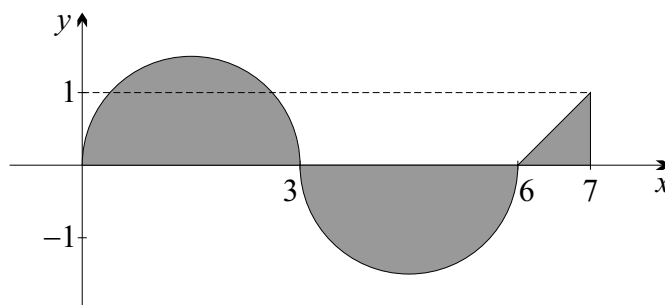
Show all necessary working.

Start a new booklet for each question.

Question Eleven	(16 marks)	Use a separate writing booklet.	Marks
(a)	Use your calculator to find $\frac{e^3}{2}$ correct to two decimal places.		1
(b)	Simplify $\frac{(e^x)^4}{e^x}$.		1
(c)	A parabola has equation $x^2 = 8y$. Find:		3
	(i) the coordinates of the vertex,		
	(ii) the coordinates of the focus,		
	(iii) the equation of the directrix.		
(d)	Differentiate:		3
	(i) $\frac{x^4}{2}$		
	(ii) $3e^{2x}$		
	(iii) $(2x - 1)^5$		
(e)	Find a primitive of:		3
	(i) $x + 16$		
	(ii) e^{4x+1}		
	(iii) $\sqrt{x}$		
(f)	Sketch on a number plane the locus of a point P which moves so that it is always 3 units from the origin. Write down the equation of the locus of P .		2

Question ELEVEN (Continued)

(g)



The function $y = f(x)$, for $0 \leq x \leq 7$, is shown above. The curves are semicircular arcs.

(i) Find $\int_0^7 f(x) dx$.

1

(ii) Find the exact total area of the shaded parts.

2**Question Twelve** (16 marks) Use a separate writing booklet.**Marks**

(a) Evaluate the following definite integrals.

(i) $\int_{-1}^1 (6x - 2) dx$

2

(ii) $\int_1^2 \frac{1}{x^3} dx$

2

(b) By completing squares, find the centre and radius of the circle $x^2 + y^2 - 4x + 8y = 5$.

2

(c) Given that $f'(x) = 2x^2 - 6$, find $f(x)$ if $f(1) = 0$.

2

(d) Consider the function $y = x^3 - 6x^2 + 7$.

(i) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

2

(ii) Find the coordinates of any stationary points and determine their nature.

2

(iii) Find the coordinates of the point of inflexion. You must show that it is a point of inflexion.

2

(iv) Sketch the graph of the function, clearly showing all stationary and inflexion points. Do NOT attempt to find any x -intercepts.

2

Question Thirteen (16 marks) Use a separate writing booklet.

Marks

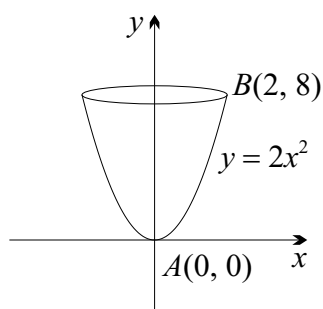
(a) Find the first and second derivatives of e^{x^2} . **3**

(b) Use the quotient rule to differentiate $y = \frac{2e^{2x+3}}{x+3}$. In your answer simplify the numerator as far as possible. **2**

(c) Use Simpson's Rule with three function values to approximate $\int_0^1 2^x dx$. Give your answer correct to two decimal places. **2**

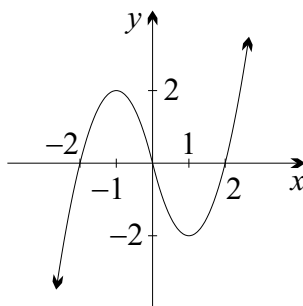
(d) Find the value of p if $\int_1^p (3x + 4) dx = 20$ and $p > 1$. **3**

(e) **3**



The diagram above shows a cup of height 8 cm whose width at the top is 4 cm. It is formed by rotating the arc AB of the parabola $y = 2x^2$ about the y -axis. Find the exact volume of the cup.

(f)



The graph of $y = f(x)$ is sketched above. Sketch on separate diagrams, clearly indicating any x -intercepts, possible graphs of:

(i) $y = f'(x)$ **2**

(ii) $y = f''(x)$ **1**

Question Fourteen (16 marks) Use a separate writing booklet.

Marks

- (a) Use the second derivative to explain why the graph of the function $y = e^{-2x}$ is always concave up. 1
- (b) Find the equation of the normal to the curve $y = x + e^x$ at the point where the curve cuts the y -axis. 3
- (c) Sketch a graph of the parabola $6x + y^2 = 18$ clearly indicating the vertex, focus, and directrix. 3
- (d) A car's velocity v in metres per second is recorded each second as it accelerates along a drag strip. The table below gives the results. 2

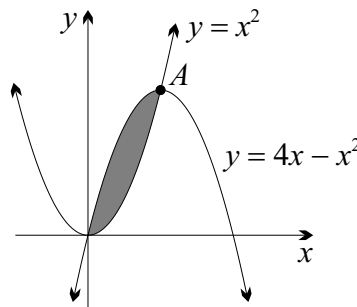
$t(s)$	0	1	2	3	4	5
$v(ms^{-1})$	0	15	31	48	64	83

Given that the distance travelled may be found by calculating the area under a velocity/time graph, use the trapezoidal rule to estimate the distance travelled by the car in the first five seconds.

- (e) Solve for x : 3

$$e^{2x} + e^x - 2 = 0$$

- (f)



The diagram shows the curves $y = x^2$ and $y = 4x - x^2$ which intersect at the origin and at point A.

- (i) Find the coordinates of point A. 2
- (ii) Hence find the area enclosed by the two parabolas. 2

Question Fifteen (16 marks) Use a separate writing booklet.**Marks**

- (a) A continuous function
- $y = f(x)$
- satisfies all of the following conditions:

3

$$f(2) > 0$$

$$f(-4) < 0$$

$$f'(x) > 0$$

$$f''(x) < 0$$

Draw a possible sketch of the function for $-4 \leq x \leq 2$.

- (b) Suppose that
- $y = e^{kx}$
- .

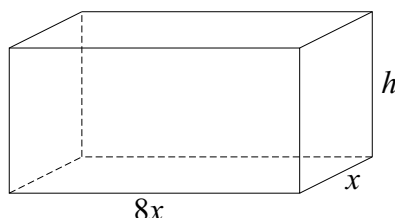
(i) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

2

(ii) Find the value of k such that $y = 2\frac{dy}{dx} - \frac{d^2y}{dx^2}$.

2

- (c)



The diagram above shows the framework of a storage container which has been constructed in the shape of a rectangular prism. The container is eight times as long as it is wide, and has breadth x metres and height h metres.

- (i) Find, in terms of
- x
- and
- h
- , an expression for the total length
- L
- of steel required to construct the frame.

2

- (ii) The container has volume
- 2304 m^3
- .

$$(\alpha) \text{ Show that } h = \frac{288}{x^2}.$$

2

$$(\beta) \text{ Show that } L = 36x + \frac{1152}{x^2}.$$

1

- $$(\gamma) \text{ Find the dimensions of the container so that the minimum length of steel is used in the construction of the frame.}$$

4

 End of Section II

END OF EXAMINATION

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

SYDNEY GRAMMAR SCHOOL



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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

CANDIDATE NUMBER:

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

Question Ten

A ☐ B ☐ C ☐ D ☐

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A ☐ B ☐ C ☒ D ☐

Question Three

A ☒ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☒ D ☐

Question Five

A ☐ B ☒ C ☐ D ☐

Question Six

A ☐ B ☐ C ☒ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☒

Question Eight

A ☒ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☒ C ☐ D ☐

Question Ten

A ☐ B ☐ C ☐ D ☒

MATHS 2-UNIT SOLUTIONS HALF-YEARLY 2012

QUESTION 11.

(a) 10.04 ✓

(b) $\frac{e^{4x}}{e^x} = e^{3x}$ ✓

(c) $x^2 = 8y$

(i) Vertex $V(0,0)$ ✓

(ii) Focus

$a=2$ $S(0,2)$ ✓

(iii) Directrix $y = -2$ ✓

(g) (i) $\int_0^7 f(x) dx = \frac{1}{2}x|x|$
 $= \frac{1}{2}$ ✓

(ii) $A = \pi \left(\frac{3}{2}\right)^2 + \frac{1}{2}$ ✓
 $= \frac{9\pi}{4} + \frac{1}{2}$ units² ✓

(d) (i) $\frac{d}{dx} \left(\frac{1}{2}x^4\right) = \frac{1}{2} \times 4x^3$

$= 2x^3$ ✓

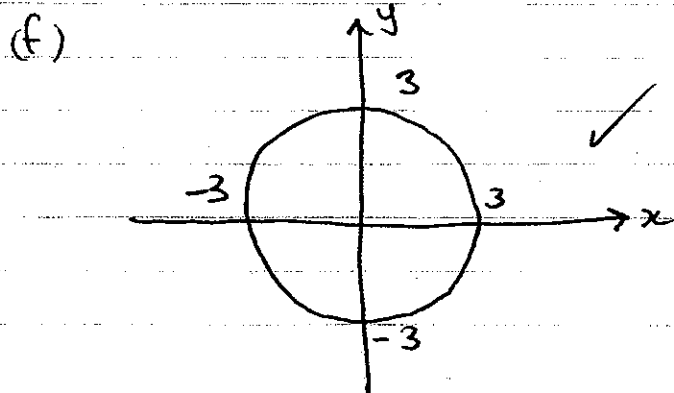
(ii) $\frac{d}{dx} (3e^{2x}) = 6e^{2x}$ ✓

(iii) $\frac{d}{dx} ((2x-1)^5) = 5 \times 2 (2x-1)^4$ ✓
 $= 10(2x-1)^4$ ✓

(e) (i) $\int x+16 dx = \frac{1}{2}x^2 + 16x + c$ ✓

(ii) $\int e^{4x+1} dx = \frac{1}{4}e^{4x+1} + c$ ✓

(iii) $\int x^{\frac{1}{2}} dx = \frac{2}{3}x^{\frac{3}{2}} + c$ ✓
 $= \frac{2x\sqrt{x}}{3} + c$



locus of P : $x^2 + y^2 = 9$ ✓

MATHS 2-UNIT SOLUTIONS

QUESTION 12

$$\begin{aligned} \text{(a) (i)} \quad \int_{-1}^1 (6x-2) dx &= \left[\frac{6x^2}{2} - 2x \right]_{-1}^1 \\ &= [3x^2 - 2x]_{-1}^1 \\ &= (3-2) - (3+2) \\ &= -4 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \int_1^2 \frac{1}{x^2} dx &= \int_1^2 x^{-2} dx \\ &= \left[\frac{x^{-1}}{-1} \right]_1^2 \\ &= \left[-\frac{1}{x} \right]_1^2 \\ &= \left(-\frac{1}{2} \right) - \left(-1 \right) \\ &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad x^2 + y^2 - 4x + 8y &= 5 \\ x^2 - 4x + 4 + y^2 + 8y + 16 &= 5 + 4 + 16 \\ (x-2)^2 + (y+4)^2 &= 25 \end{aligned}$$

Centre (2, -4)
Radius 5 units

$$\begin{aligned} \text{(c)} \quad f'(x) &= 2x^2 - 6 \\ f(x) &= \frac{2x^3}{3} - 6x + C \end{aligned}$$

$$\text{But } f(1) = 0$$

$$0 = \frac{2}{3} - 6 + C$$

$$C = \frac{16}{3}$$

$$\therefore f(x) = \frac{2x^3}{3} - 6x + \frac{16}{3}$$

$$\text{(d)} \quad y = x^3 - 6x^2 + 7$$

$$\text{(i)} \quad \frac{dy}{dx} = 3x^2 - 12x$$

$$\frac{d^2y}{dx^2} = 6x - 12$$

$$\text{(ii) Stationary points when } \frac{dy}{dx} = 0$$

$$3x^2 - 12x = 0$$

$$3x(x-4) = 0$$

$$\therefore x = 0 \text{ or } 4$$

$$\text{When } x=0 \quad y=7 \text{ and } \frac{d^2y}{dx^2} = -12 < 0$$

So (0, 7) is a local maximum turning pt

$$\text{When } x=4 \quad y = 4^3 - 6 \times 4^2 + 7 = -25$$

$$\frac{d^2y}{dx^2} = 12$$

> 0 So (4, -25) is a local minimum turning point

$$\text{(iii) Possible pt. of inflexion when } \frac{d^2y}{dx^2} = 0$$

$$6x - 12 = 0$$

$$x = 2$$

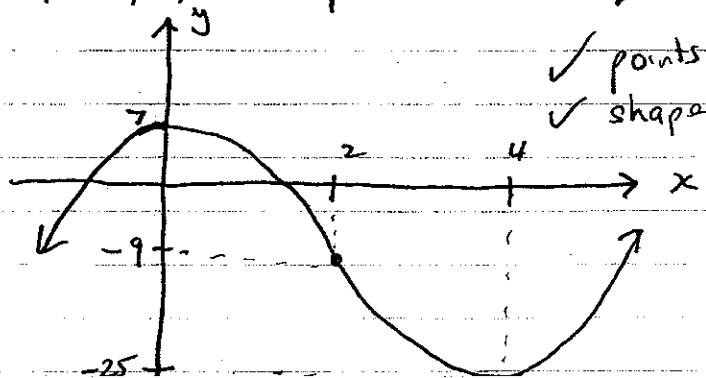
$$\text{When } x=2 \quad y = 2^3 - 6 \times 2^2 + 7 = -9$$

Check concavity changes

x	0	2	4
$\frac{d^2y}{dx^2}$	-12	0	12
	↙	↘	↗

So, (2, -9) is a point of inflexion

(iv)



QUESTION 13

(a) $y = e^{x^2}$
 $\frac{dy}{dx} = 2xe^{x^2}$ ✓
 $\frac{d^2y}{dx^2} = 2x \cdot 2xe^{x^2} + 2e^{x^2}$ ✓
 $= 4x^2e^{x^2} + 2e^{x^2}$ ✓
 $= 2e^{x^2}(2x^2 + 1)$ ✓

(b) $y' = \frac{(x+3)4e^{2x+3} - 2e^{2x+3}}{(x+3)^2}$ ✓
 $= \frac{e^{2x+3}(4(x+3) - 2)}{(x+3)^2}$ ✓
 $= \frac{e^{2x+3}(4x+10)}{(x+3)^2}$ ✓

(c)

x	0	$\frac{1}{2}$	1
$f(x)$	1	$\sqrt{2}$	2

$\int_0^1 2^x \cdot dx \doteq \frac{1-0}{6} (1 + 4\sqrt{2} + 2)$ ✓
 $\doteq \frac{1}{6} (3 + 4\sqrt{2})$ ✓
 $\doteq 1.44$ ✓

(d) $\int_1^p (3x+4) dx = \left[\frac{3x^2}{2} + 4x \right]_1^p$ ✓
 $= 20$

$\frac{3p^2}{2} + 4p - \left(\frac{3}{2} + 4 \right) = 20$

$\frac{3p^2}{2} + 4p - \frac{11}{2} = 20$

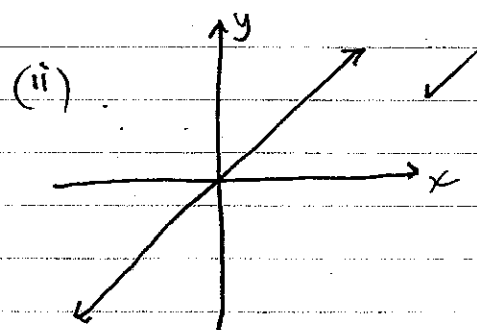
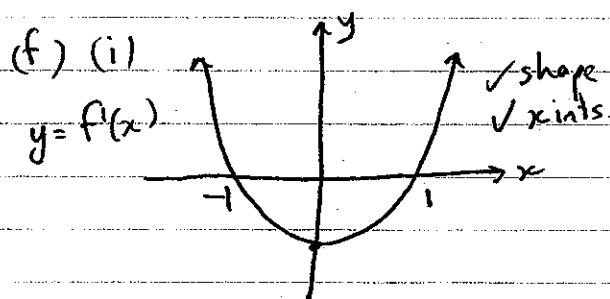
$3p^2 + 8p - 11 = 40$ ✓

$3p^2 + 8p - 51 = 0$

$(3p+17)(p-3) = 0$

$\therefore p = 3 \quad (p > 1)$ ✓

(e) $V = \pi \int_0^8 x^2 dy$
 $= \pi \int_0^8 \frac{y}{2} dy$ ✓
 $= \frac{\pi}{2} \int_0^8 y dx$
 $= \frac{\pi}{2} \left[\frac{y^2}{2} \right]_0^8$ ✓
 $= \frac{64\pi}{4}$
 $= 16\pi \text{ units}^3$ ✓



QUESTION 14

(a) $y = e^{-2x}$ $y' = -2e^{-2x}$ $y'' = 4e^{-2x}$

$\frac{4}{e^{2x}} > 0$ for all x , so $y = e^{-2x}$ is always concave up.

(b) $y = x + e^x$

$\frac{dy}{dx} = 1 + e^x$

At $x = 0$

$\frac{dy}{dx} = 1 + 1 = 2$

So gradient of tangent is 2.
and $\therefore$ gradient of normal is $-\frac{1}{2}$

When $x = 0$

$y = 0 + e^0 = 1$

$y - y_1 = m(x - x_1)$

$y - 1 = -\frac{1}{2}(x - 0)$

$y - 1 = -\frac{1}{2}x$

$2y - 2 = -x$

$x + 2y - 2 = 0$

(c) $y^2 = -6x + 18$

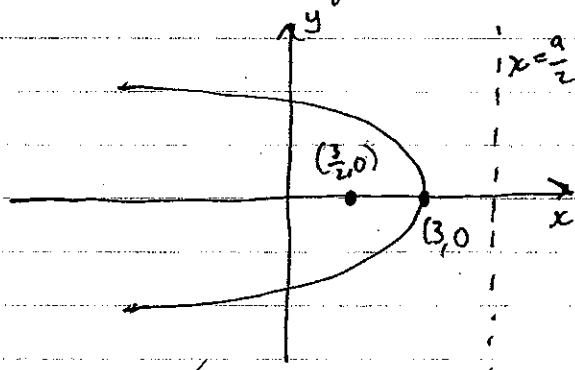
$y^2 = -6(x - 3)$

$a = \frac{1}{2}$

Vertex $(3, 0)$

Focus $(\frac{3}{2}, 0)$

Directrix $x = \frac{9}{2}$



(d) $D = \frac{1}{2}(0 + 2(15) + 2(31) + 2(48) + 2(64) + 83)$

$= \frac{399}{2}$

$= 199.5 \text{ Metres}$

(e) $e^{2x} + e^x - 2 = 0$

$(e^x - 1)(e^x + 2) = 0$

Either $e^x = 1$

$x = 0$

or $e^x = -2$

No solution, $e^x > 0$

(f) (i) $x^2 = 4x - x^2$

$2x^2 - 4x = 0$

$2x(x - 2) = 0$

$x = 0 \text{ or } 2$

When $x = 2$, $y = 4$

So A has coordinates $(2, 4)$

(ii) $\int_0^2 (4x - x^2 - x^2) dx$

$= \int_0^2 (4x - 2x^2) dx$

$= \left[\frac{4x^2}{2} - \frac{2x^3}{3} \right]_0^2$

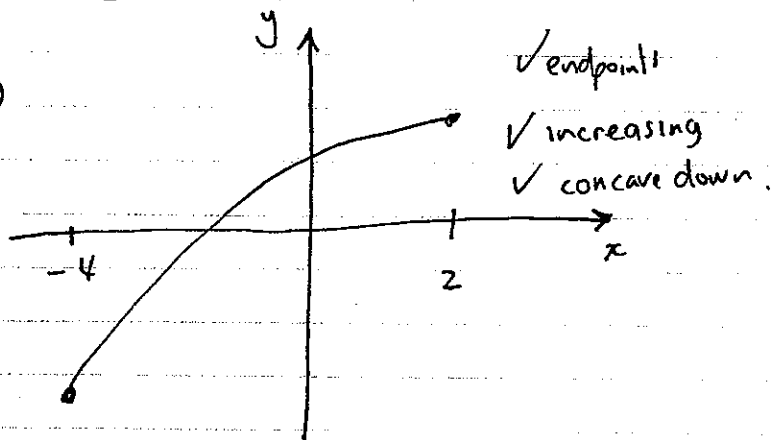
$= \left[2x^2 - \frac{2x^3}{3} \right]_0^2$

$= \left(8 - \frac{16}{3} \right) - 0$

$= \frac{8}{3} \text{ units}^2$

Question 15

(a)



(b) (i) $\frac{dy}{dx} = k e^{kx}$ ✓

$\frac{d^2y}{dx^2} = k^2 e^{kx}$ ✓

(ii) $e^{kx} = 2k e^{kx} - k^2 e^{kx}$
 $k^2 e^{kx} - 2k e^{kx} + e^{kx} = 0$
 $e^{kx} (k^2 - 2k + 1) = 0$ ✓
 $e^{kx} (k-1)^2 = 0$

When $e^{kx} = 0$

No solution

When $(k-1) = 0$ ✓
 $k = 1$

(c) (i) $L = 4(8x + x + h)$ ✓
 $= 36x + 4h$ ✓

(ii) (a) $V = 8x \times x \times h$
 $= 8x^2 h$
 $2304 = 8x^2 h$ ✓
 $h = \frac{2304}{8x^2}$
 $= \frac{288}{x^2}$ ✓

(b) $L = 36x + 4h$
 $= 36x + 4 \times \frac{288}{x^2}$

$L = 36x + \frac{1152}{x^2}$ ✓

(c) Minimum occurs when

$\frac{dL}{dx} = 0$

✓ $36 - \frac{2304}{x^3} = 0$

$x^3 = \frac{2304}{36}$

$x^3 = 64$

So $x = 4$ ✓

Check that this is a minimum

$\frac{d^2L}{dx^2} = \frac{6912}{x^4}$ ✓
 > 0

So when $x = 4$, $h = \frac{288}{16}$
 $= 18$

(is a minimum)

Dimensions of container
 $4m \times 32m \times 18m$ ✓